

## WEEK 2:

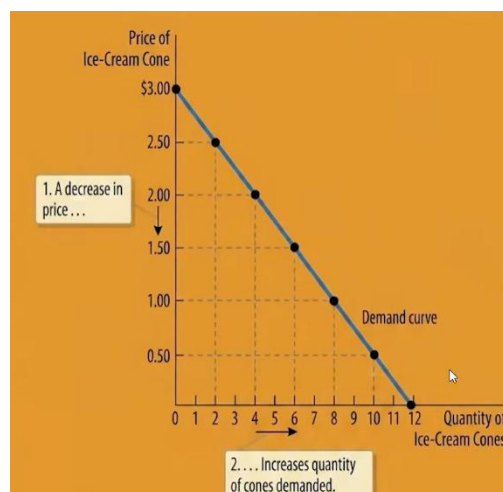
### Market:

- A group of buyers and sellers of a particular good or service.
- Buyers → demand ; sellers → supply
  - Organized market → homogenous place, where buyers and sellers meet at a specific time and place.
  - Unorganized market → sellers are scattered – offer slightly different products, and buyers choose where to purchase.
  - Competitive market → many buyers and sellers. Price is determined by the market as a whole – buyers have negligible impact on the price.
  - Non-competitive market
    - Monopoly : a single seller sets the price.
    - Oligopoly: market dominated by a small number of large firms.

### Demand:

- Quantity of goods/services that consumers are **willing** and **able** to buy at different prices during a given period of time.
- Demand has to be backed by purchasing power.
- Law of demand:
  - When the price of a good rises, the quantity demanded of the good falls; when the price falls, the quantity demanded rises.
- Demand schedule → table that displays the relationship between the price of a good and the quantity demanded
- Demand curve → graphical representation of the demand schedule.  
It slopes downward because a lower price naturally increases the quantity

| Price of Ice-Cream Cone | Quantity of Cones Demanded |
|-------------------------|----------------------------|
| 0                       | 12                         |
| 0.5                     | 10                         |
| 1                       | 8                          |
| 1.5                     | 6                          |
| 2                       | 4                          |
| 2.5                     | 2                          |
| 3                       | 0                          |

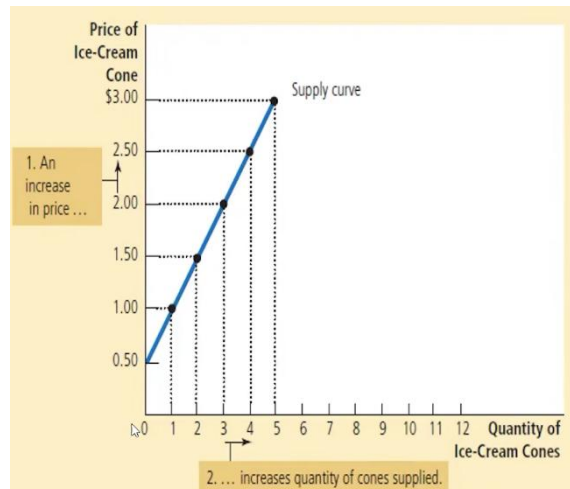


- Market demand → sum of all individual demands for a product.
- Quantity demand → sum of quantities demanded by the buyers at each price

## Supply:

- The quantity of a good or service that producers are willing and able to offer for sale at different prices during a given period of time.
- Law of supply:
  - When price of a good rises, the quantity supplied also rises; when price falls quantity supplied falls.
- Supply schedule: relationship bw price and quantity
- Supply curve: plotting supply schedule
- Market supply: aggregate of all individual sellers in the market

| Price of Icecream Cone | Quantity of Cones Supplied |
|------------------------|----------------------------|
| 0                      | 0                          |
| 0.5                    | 0                          |
| 1                      | 1                          |
| 1.5                    | 2                          |
| 2                      | 3                          |
| 2.5                    | 4                          |
| 3                      | 5                          |



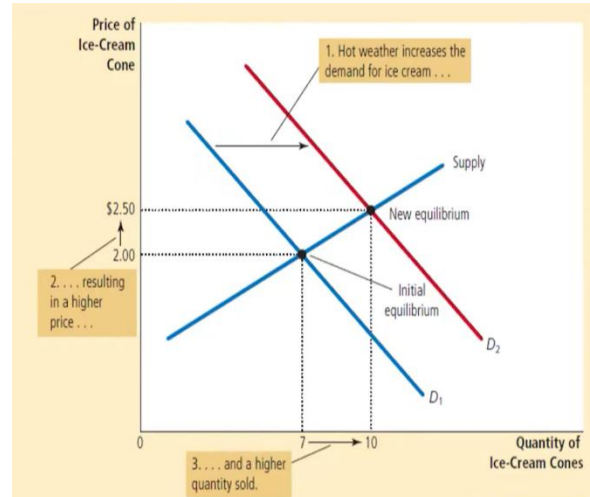
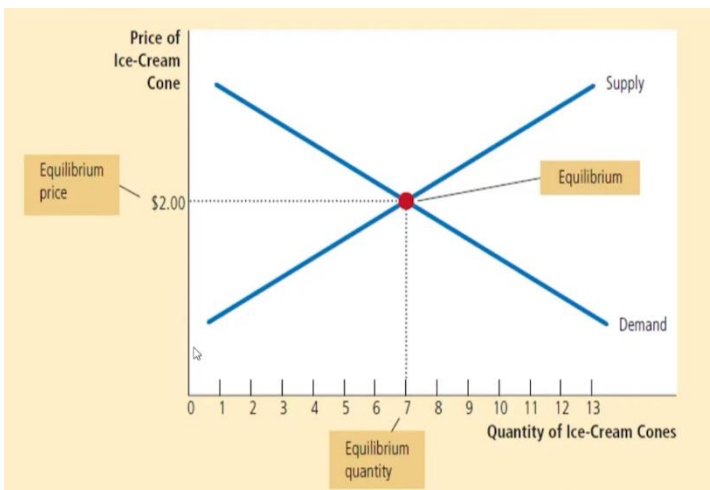
## Shifts in Demand Curve:

- Increase in Quantity demanded → increase in demand
- Decrease in quantity demanded → decrease in demand
- Key variables:
  - Income: higher income increases demand for normal goods.
    - If demand rises when income falls, it's an inferior good.
  - Price of related goods:
    - Two goods are substitutes if a fall in price of one reduces the demand of the other
    - Two goods are complements if a fall in price of one increases the demand of the other
  - Preference/ tastes:
    - Positive preferences increase demand
  - Future expectations
  - Number of buyers: increase in the number of consumers increases market demand.

## Shift in Supply curve

- Key variables:
  - Input prices: if input prices rise, production becomes less profitable
  - Technology: advancement in tech reduces labour costs and increases supply of good

- Expectations: if a firm expects market price to rise, it might store some of its current production, reducing supply available to market
- Number of sellers: if sellers leave the business, market supply will decrease.
- Equilibrium: Price has reached level where quantity supplied = quantity demanded.
- Equilibrium price: price that balances quantity supplied and demanded
- Equilibrium quantity: quantity supplied and demanded at equilibrium
  - surplus: market price above equilibrium price → surplus occurs because quantity supplied > demanded
  - shortage: quantity supplied < demanded



### Elasticity of demand:

- Elasticity: degree of responsiveness of one variable to changes in another
- Price of elasticity demanded: measures how much quantity demanded changes as a response to a change in price

### Example:

- If Netflix raises prices and many customers cancel subscriptions: **demand is elastic**
- If petrol prices increase but people still buy almost the same amount: **demand is inelastic**

$$E_D = \frac{\% \text{ change in Demand of Goods}}{\% \text{ change in Price of Goods}}$$

$$E_D = \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100}$$

$$E_D = \left(\frac{P}{Q}\right) \left(\frac{\Delta Q}{\Delta P}\right)$$

### Determinants of elasticity:

- Availability of close substitutes
- Definition of markets: narrowly defined markets are more elastic → easier to find close substitutes
- Necessities vs luxury: necessities are inelastic. Luxury is elastic

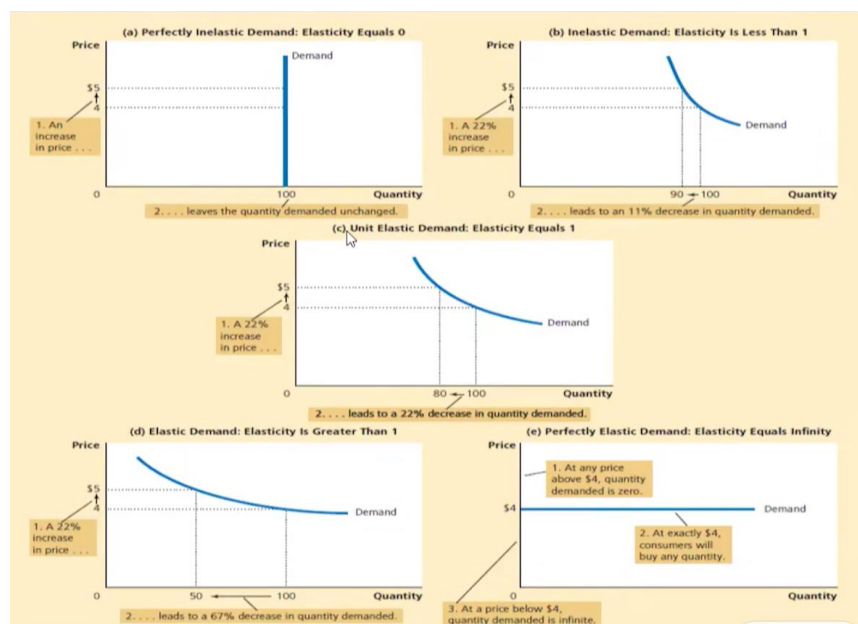
## Time horizon:

- Goods tend to have more elastic demand over a longer time horizon.

## Price elasticity of demand:

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

| Type of Demand             | Meaning                                                                   | Example                                              |
|----------------------------|---------------------------------------------------------------------------|------------------------------------------------------|
| Elastic Demand             | A small change in price leads to a large change in quantity demanded      | Restaurant meals, branded clothing                   |
| Inelastic Demand           | A change in price leads to a relatively small change in quantity demanded | Petrol, medicines                                    |
| Unit Elastic Demand        | Percentage change in demand equals percentage change in price             | A 10% fall in price leads to a 10% rise in demand    |
| Perfectly Elastic Demand   | Even a tiny increase in price causes demand to fall to zero               | Identical products in a perfectly competitive market |
| Perfectly Inelastic Demand | Demand remains constant regardless of price changes                       | Life-saving insulin for certain patients             |

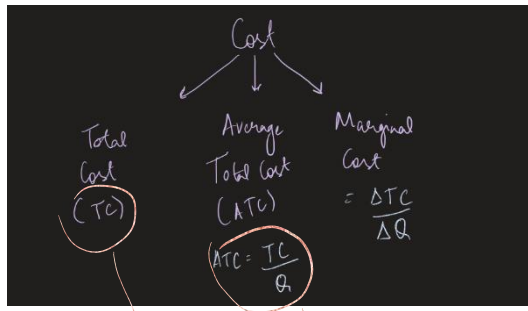


$$\text{Total revenue} = P \times Q$$

$P \rightarrow$  price of good

$Q \rightarrow$  quantity sold

- **Elastic demand:** decrease in revenue
- **Inelastic demand:** increase in revenue
- **Unit elastic demand:** revenue remains constant



$\Rightarrow TC/Q$   
 $AC = AFC + AVC$   
 average fixed cost      average variable cost

$TC = TFC + TVC$   
 total fixed cost      total variable cost

$AFC = \frac{FC \text{ (fixed cost)}}{Q}$

| Output<br>( $Q$ ) | TFC<br>(given) | TVC<br>(given) | TC<br>(TFC+TVC) | AFC<br>$TFC/Q$ | AVC<br>$TVC/Q$ | MC<br>$\Delta TC / \Delta Q$ | AC<br>(AFC+AVC) } also $\frac{TC}{Q}$ |
|-------------------|----------------|----------------|-----------------|----------------|----------------|------------------------------|---------------------------------------|
| 0                 | 20             | 0              | 20              | (20/0) ×       | —              | —                            | —                                     |
| 1                 | 20             | 10             | 30              | 20             | 10             | $(30-20)/1-0 = 10$           | 30                                    |
| 2                 | 20             | 18             | 38              | 10             | 9              | $(38-30)/2-1 = 8$            | 19                                    |

→ This is how to calc. all the values given  $Q$ , TFC, TVC.

Total Revenue = Profit + Cost  
(TR)

Economic Profit = TR - (Explicit Cost + Implicit Cost)  
(EP)      Opportunity Cost

Accounting Profit = TR - Explicit Cost

① EP = 0 → Stable (Break even condition)

② EP → +ve : Expand ; Reduce price

③ EP → -ve : exit from market



A company operating a Software-as-a-Service (SaaS) platform experiences a massive surge in growth driven by increasing **online education, e-commerce, and other digital services along with digital money transaction**. While their **Total Fixed Costs** (such as office rent, core system architecture, and baseline salaries) **remain unchanged**, they are able to distribute these overhead expenses across millions of new subscribers. Which of the following trends is most likely to occur?

- Options
- ☐ A. The Average Fixed Cost (AFC) per subscriber will continuously decline toward zero.
  - ☐ B. The Marginal Cost (MC) will become exactly equal to the Total Fixed Cost.
  - ☐ C. The Average Total Cost (ATC) will begin to increase immediately.
  - ☐ D. The Total Variable Cost (TVC) will automatically drop to zero.

Question 1 / 10 Flag for review 2 marks

You purchased non-refundable plane tickets for a trip to Mumbai this weekend for ₹5,500/-. You estimate that the trip would give you ₹10,000/- worth of value (enjoyment and benefits). Your friend has just called and informed you that he won 2 free pavilion tickets to the IPL final match between Gujarat Titans and RCB (cost ₹8,000/- each) and wants to know if you would like to join him. Calculate the effective value one must gain from the IPL final to justify missing the scheduled flight to Mumbai.

- Options
- ☐ A. ₹5500
  - ☐ B. ₹16000
  - ☒ C. ₹10000
  - ☐ D. ₹8000
- Clear Selection

A chip manufacturing company, working on developing hardware for enterprises calculates that at its current output of 10,000 microchips, its Average Total Cost (ATC) is ₹1500.00 per chip. The Marginal Cost (MC) of producing the 10,001<sup>st</sup> chip is ₹800.00. What will happen to the Average Total Cost if the firm decides to produce that additional chip?

- Options
- ☒ A. The ATC will decrease.

- **Average Fixed Cost (AFC) = Total Fixed Cost (TFC) ÷ Quantity (Q).**
- If TFC remains unchanged and the number of subscribers grows into the millions, AFC keeps falling because the same fixed costs are spread over more users.
- As *Q* becomes very large, AFC approaches **zero**.

Why the others are wrong:

- **B:** Marginal Cost is the cost of serving one additional subscriber; it does not become equal to Total Fixed Cost.
- **C:** ATC does not necessarily increase. In fact, spreading fixed costs over more subscribers tends to **reduce** ATC, at least initially.
- **D:** Variable costs (bandwidth, customer support, cloud usage, etc.) do not automatically become zero.

**Answer: A. The Average Fixed Cost (AFC) per subscriber will continuously decline toward zero.**

This is a classic **sunk cost** problem.

- You already purchased the Mumbai flight for **₹5,500**, and it is **non-refundable**.
- The Mumbai trip is expected to provide **₹10,000** of value.
- Whether you go to Mumbai or not, the **₹5,500 is gone**. Therefore, it should not affect the decision.

To justify skipping Mumbai for the IPL final, compare the value of the alternatives going forward:

- Value of going to Mumbai = **₹10,000**
- Value of attending IPL final = ?

You should choose the IPL final only if its value exceeds the value of the Mumbai trip.

Therefore, the **minimum effective value required from attending the IPL final is:**

₹10,000

The fact that the IPL tickets are worth ₹8,000 each is largely irrelevant unless you could sell them. If the tickets are non-transferable and the goal is to maximize personal enjoyment/benefit, the IPL final must provide **more than ₹10,000 of value** to justify missing Mumbai.

**Answer: ₹10,000 (break-even value); anything above ₹10,000 makes skipping Mumbai worthwhile.**

★ **The intuition:**

**ATC behaves like a “class average.”**

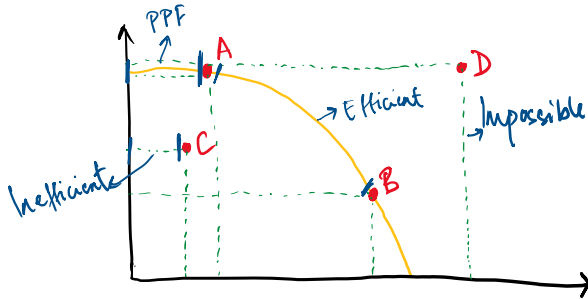
**MC behaves like the “next student’s score.”**

If the next student scores **below** the current average → the average **falls**.  
If the next student scores **above** the current average → the average **rises**.

Economics uses the **same logic**.

## PPF – Production Possibility Frontier

- Graphs the maximum possibility combination of two goods an economy can produce when all resources are fully and efficiently utilized.
- Points on the curve → represents productive efficiency
- Points inside the curve → represents inefficiency of resources (wasted materials)
- Points outside the curve → resources currently unattainable

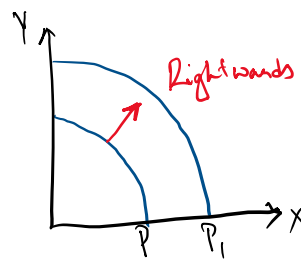


- As you shift resources from one good to another you must give up increasingly larger amounts of the first good because resources are not equally suited for producing both goods
- Assumptions:
  - Input fixed
  - Resources are fully utilised
  - Technology and other conditions are fixed
- Marginal opportunity cost (MOC) =  $\frac{\Delta S}{\Delta G} \rightarrow \text{Change in sacrifices} \quad (\Delta Y / \Delta X)$   
 $\Delta G \rightarrow \text{Change in gain}$

### Shifts in PPF:

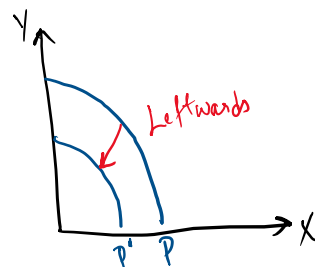
#### 1. Rightward shift

Due to advancement/ tech upgrade



#### 2. Leftward shift:

Due to insufficient resources



**Rotation in PPF:**

- Fixing the y point – changing the x point

**PPF:** Concept of efficiency → output/input

- Efficiency > 100% → over utilisation of resources
- Efficiency = 100% → no wastage of resources
- Efficiency < 100% → underutilisation of resources

